



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Software Concepts!



Learning Outcome:

Understands systems theories, business processes and basic concepts on IT.

Create a map of processes that allows identifying areas in an information system, using IT tools to generate reports based on the company's needs.

Can Do Statements:

- I can list the most important concepts about software engineering.
- I can give brief information about software engineering concepts.
- I can explain briefly what their importance in software engineering is.

Ask your partner:

- How do you think websites or social networks identify your password and give you access?
- Do you think using a data base for a business is an advantage? Why? Why not?





Let's Explore!

Get ready to be a software engineer!



Instructions: Read the text below.

Teacher: Let's discuss some software engineering concepts. So, it's time for you to ask questions.

Student: I have a list of the main concepts. These include relational databases, security, cloud computing, concurrency, caching, and hashing. Those aren't clear enough to me. Could you give me a brief definition of them, please?

Teacher: Yes, those are just six. Let's start with relational databases. These are collections of data items that are organized as a group of tables from which you can access data. The security protocol is also important because a person or company can exchange and protect confidential information. In general, internet security supports authentication, authorization, and information transmission that is often encrypted via SSL.

Student: I remember that SSL code means Secure Socket Layer. SSL is a process that we use to recognize passwords in a website.

Teacher: Yes, that's true. Cloud computing is another useful and popular concept these days. With this, you can store and access data and programs over the Internet without the necessity of using your computer's hard drive.

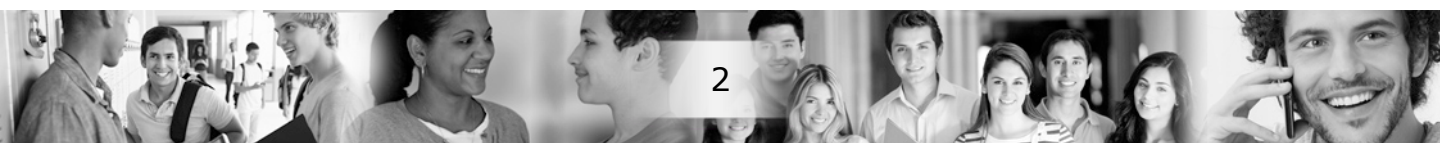
Student: I think Microsoft Azure is a good example or if. This program uses a platform as a service, and it allows cloud computing.

Teacher: Correct. We also have concurrency. Concurrency is the ability of a database to allow multiple users to make multiple transactions simultaneously. This is one of the main properties that differentiates a database, such as the Java library, from other types of data storage, such as spreadsheets. Now, can you tell what caching and hashing are?

Student: Caching comes from 'cache'. The cache is a part of the local internal memory that helps us to store information from database subcategories. Hashing, for its part, is a programming function that we use to find and place information into large files.

Teacher: Good. So, after studying these concepts, why do you think software engineering and these concepts are so important?

Student: Well, software engineering is important because it aims to satisfy the needs of the users successfully. It offers them the possibility to control their networks in the most efficient, effective, and easiest way. Managing these concepts reduces complexity, make processes safe, and it helps companies to increase their productivity!





Let's Practice!

Do you remember the main concepts of software engineering?

Instructions: Use the word(s) from the box to correctly complete the sentence.

- | | | | | | |
|----|------------------------------------|----|------------------------------------|----|------------------------------------|
| 1. | <input type="text" value="these"/> | 2. | <input type="text" value="these"/> | 3. | <input type="text" value="this"/> |
| | <input type="text" value="those"/> | | <input type="text" value="that"/> | | <input type="text" value="those"/> |
| | <input type="text" value="those"/> | | <input type="text" value="this"/> | | <input type="text" value="that"/> |

Teacher: Please, Mr. Sanz, make a list with the main concepts of software engineering.

Student: Ok, I remember three concepts, (1) _____ are relational data bases, security, and cloud computing.

Teacher: You are right. (2) _____ concepts are really important for software. What about concurrency?

Student: Yes, (3) _____ is part of the concepts too.



Let's Practice!

These pieces of information help us to understand software engineering concepts.

Instructions: Select the correct answer to complete the sentences.

- (Concurrency / Cloud computing / Relational databases) are a collection of data items that are organized in (software / clouds / tables) that (contains / contain / containing) categories.
- (Caching / Cloud computing / Concurrency) is a part of a device's memory that (stores / look for / store) the information you use. It is useful because it (saves / save / waste) the user time



Extension Activity

Make a poster explaining different software engineering concepts.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about main software engineering concepts.
- Describe what each concept is about.
- Write what the importance of those concepts for software engineering is.
- Share your poster with the class and discuss about other main concepts.

Self-Assessment

Yes

No

- I find information about main software engineering concepts. ☐ ☐
- I describe what each concept is about. ☐ ☐
- I understand what the importance of those concepts is. ☐ ☐



Target Words

- **encrypted** (adjective): something encoded, normally sent as a message
e.g. Internet security supports authentication, authorization, and information transmission that is often encrypted via SSL.
- **store** (verb): to save or put in storage
e.g. The cache is a part of the local internal memory that helps us to store information from database subcategories.
- **simultaneously** (adverb): at the same time as
e.g. Concurrency is the ability of a database to allow multiple users to make multiple transactions simultaneously.
- **socket** (noun): part of a two-way communication
e.g. I remember that SSL code means Secure Socket Layer. SSL is a process that we use to recognize passwords in a website.
- **cache** (noun): locally stored information, used to help speed things up
e.g. The cache is a part of the local internal memory that helps us to store information from database subcategories.



Learning Points

Demonstratives as pronouns and adjectives

You use the *demonstratives* as pronouns to express the distance between the speaker and the noun. Demonstrative pronouns are those that identify or point at things and persons. In addition, you can use demonstratives as adjectives to give a characteristic to a noun.

The demonstrative pronouns and demonstrative adjectives use the same words, but they differ in the position within a sentence and the intended meaning.

	NEAR	FAR
SINGULAR	<i>This</i>	<i>That</i>
PLURAL	<i>These</i>	<i>Those</i>

There are singular and plural demonstratives that you can use according to the distance: if it is near or far from your position. Look at the following examples:

*You need to store the information of **this** document in the cloud.*
(the document is near the speaker.)

***That** provides information to different users.*
(The program or application is far from the speaker.)

Use demonstrative pronouns to identify and replace nouns. In order to replace nouns, you have to make sure to leave the context clear so that there are no ambiguities. Look at the following examples:

***The records** are easy to find with the database. = **These** are easy to find with the use of query language.*

Clouds provide application software; this is an advantage.

*Web systems works with **a cache**. This is an in-memory store that contains information.*

On the contrary, the **demonstrative adjectives** modify the noun and help us indicate the position of something related to ours.

***This technique** is effective to store information.
That internal memory stores a subset of information.
These concurrency is part of application processes.
Those devices are associated to the Cloud Computing.*

So, the difference between these demonstratives is that the demonstrative adjectives are always placed before a noun while the demonstrative pronouns are placed before a verb.



Learning Points

Examples

- *This is a kind of internal memory that helps to store information.*
- *These are useful to organize and represent large data in records.*
- *That is a type of internet computing service to give resources to a group of devices.*
- *That SSL code means secure socket layer.*

Software Engineering: Main Concepts

Software engineering involves different and multiple processes for software production. There are common concepts to test, implement and develop a software, let's have a look at them.

Concept	What is it?	Common uses and examples
Relational Databases	A collection of data items. These ones are organized in tables that contain information. You can access information or put it back together without the necessity of organizing the tables.	• Order management systems • Corporate Data bases: business, to register profits and losses.
Security	It contains the verification of the user's identity or authentication, the permissions or authorizations, and the information transmission.	• Website browsers: banks transactions, Firefox browser • OAuth protocol to enable workers of a company to access information.
Cloud Computing	It is a type of Internet-based computing service that gives a group of computers and similar devices the resources to process information. It processes data from different companies.	• Infrastructure as a service (IaaS): Xen, Oracle VirtualBox, Oracle VM. • Platform as a service (PaaS): Microsoft Azure and Google App Engine. • Software as a service (SaaS): customers get access to application software and databases through a common cloud.



Learning Points

Concurrency	It has to do with parallelism inside the application. It is a group of techniques that permit a program to run different tasks simultaneously.	Java implements concurrency using different threads, which are placeholders of information associated with a single use of a program that can manage multiple concurrent users. For example, the concurrency library of core Java.
Caching	It is an internal memory that saves subcategories of information. These are commonly stored in a database. It helps to generate results based on information from the database at a cheaper price.	To distribute information, modern web applications, such as Facebook, work with the caching system called Memcached.
Hashing	It helps to the fast access to data. It facilitates the search of data organized in a sequence of large files. It is also used to place data in a cloud system.	Hash functions are used in Google's indexing service; each URL is hashed to a particular computer.